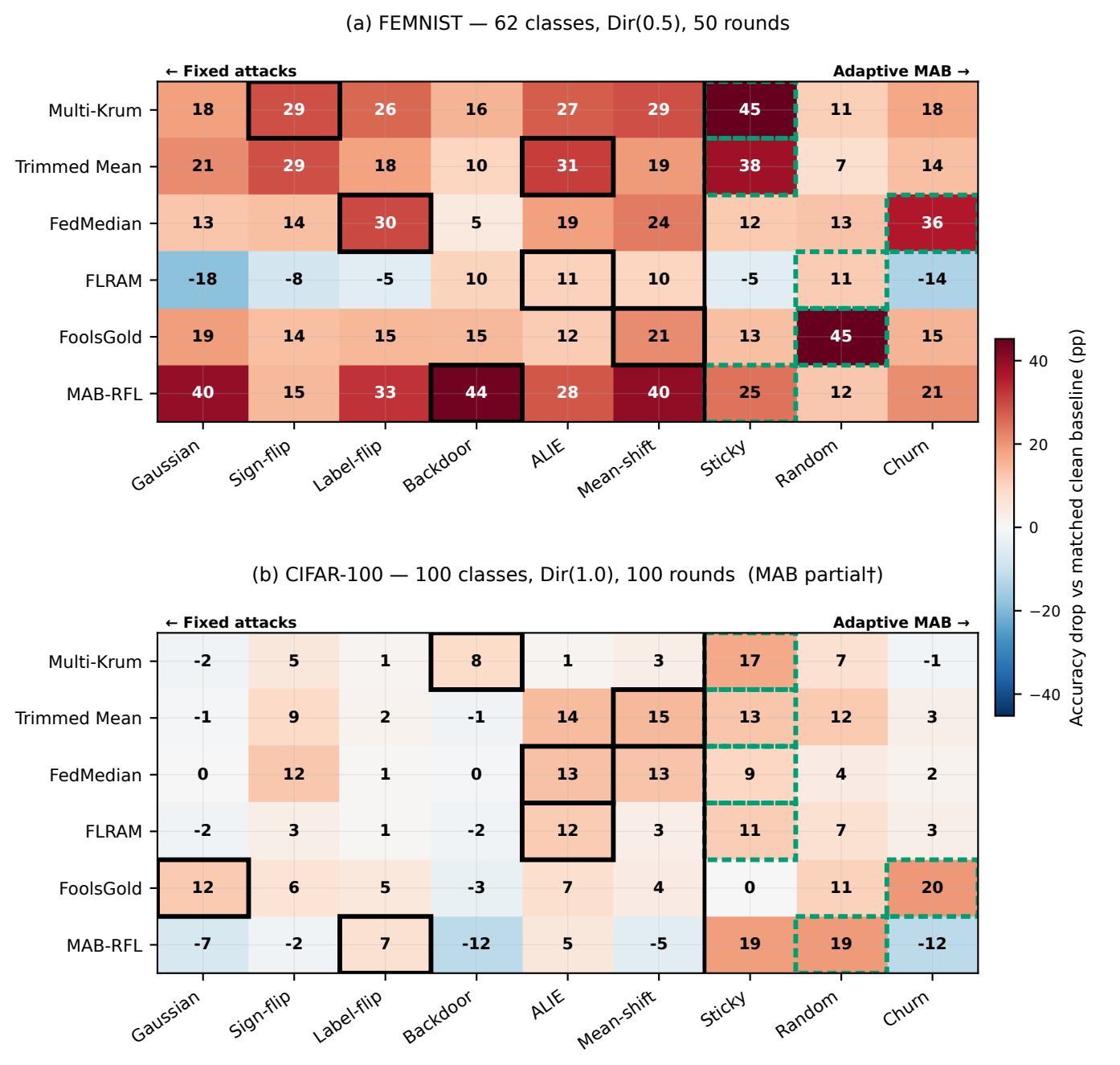
(a) FEMNIST — 62 classes, Dir(0.5), 50 rounds
← Fixed attacks
Adaptive MAB →
Multi-Krum 18 29 26 16 27 29 45 11 18
Trimmed Mean 21 29 18 10 31 19 38 7 14
FedMedian 13 14 30 5 19 24 12 13 36
FLRAM -18 -8 -5 10 11 10 -5 11 -14
FoolsGold 19 14 15 15 12 21 13 45 15
MAB-RFL 40 15 33 44 28 40 25 12 21
Gaussian
Sign-flip
Label-flip
Backdoor
ALIE
Mean-shift
Sticky
Random
Churn
(b) CIFAR-100 — 100 classes, Dir(1.0), 100 rounds (MAB partial†)
← Fixed attacks
Adaptive MAB →
Multi-Krum -2 5 1 8 1 3 17 7 -1
Trimmed Mean -1 9 2 -1 14 15 13 12 3
FedMedian 0 12 1 0 13 13 9 4 2
FLRAM -2 3 1 -2 12 3 11 7 3
FoolsGold 12 6 5 -3 7 4 0 11 20
MAB-RFL -7 -2 7 -12 5 -5 19 19 -12
Gaussian
Sign-flip
Label-flip
Backdoor
ALIE
Mean-shift
Sticky
Random
Churn
40
20
0
−20
−40
Accuracy drop vs matched clean baseline (pp)
† Gray = MAB in progress. Black box = strongest fixed attack; dashed green = strongest MAB schedule. Negative pp = attacked run exceeded its clean baseline (seed artifact).